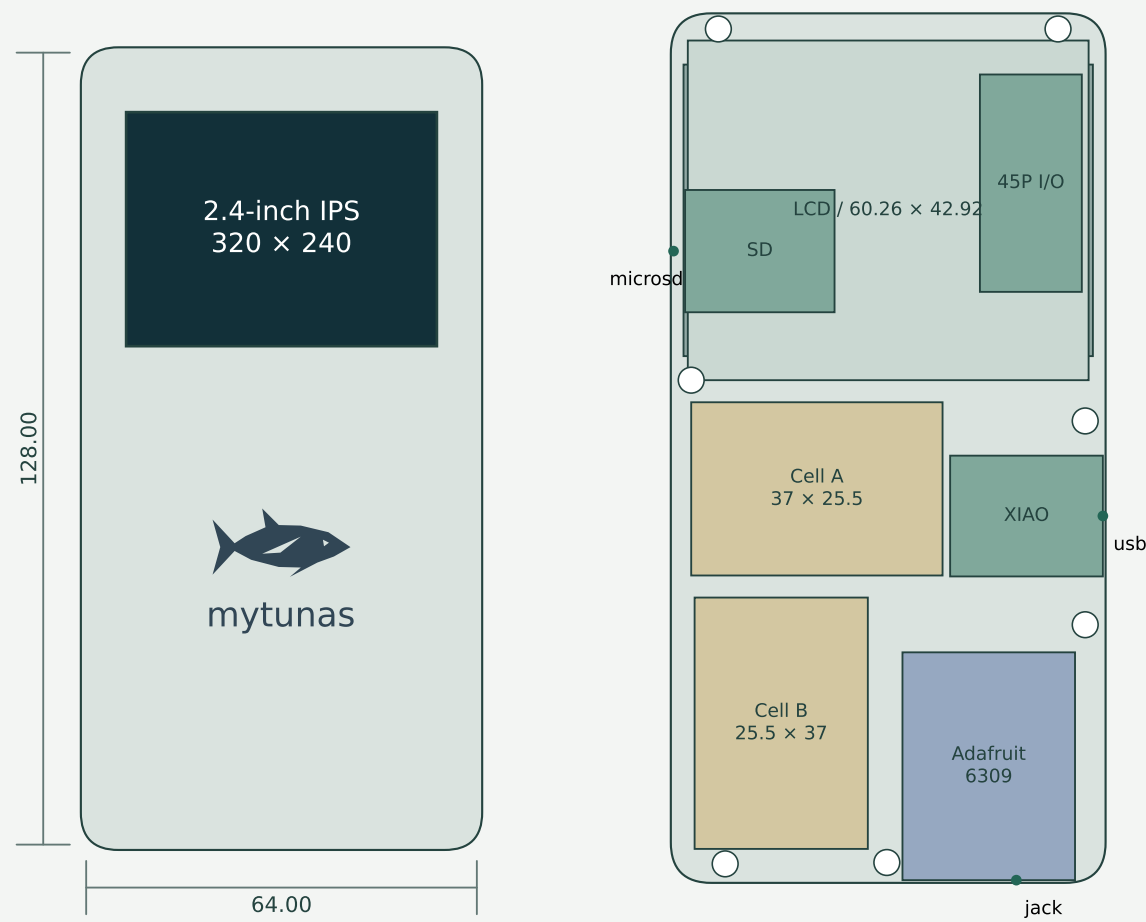


mytunas / Model 02

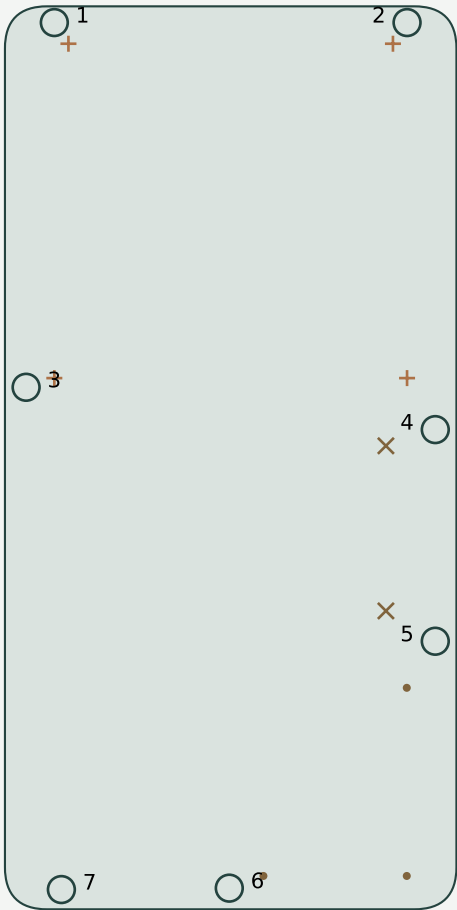
01 — Exterior and internal arrangement. Front-coordinate plan; component rectangles show validated bounds.



Feature	Dimension
Body	64 × 128 × 8.30
Corner radius	R6.00
Steel faces	0.40 each
Frame perimeter	Z0.55 to 7.75
Rear hard stops	Z7.90 contact plane
Display opening	49.56 × 37.32
Display centre	X0, Y35
USB opening	9.24 × 3.51 / rim profile
Headphone opening	Diameter 5.00
SD opening	11.50 × 1.50
Covers	Flush / removable prototype

Hole and port coordinates

02 — Datum: body centre X0/Y0; front exterior Z0. All dimensions in millimetres.

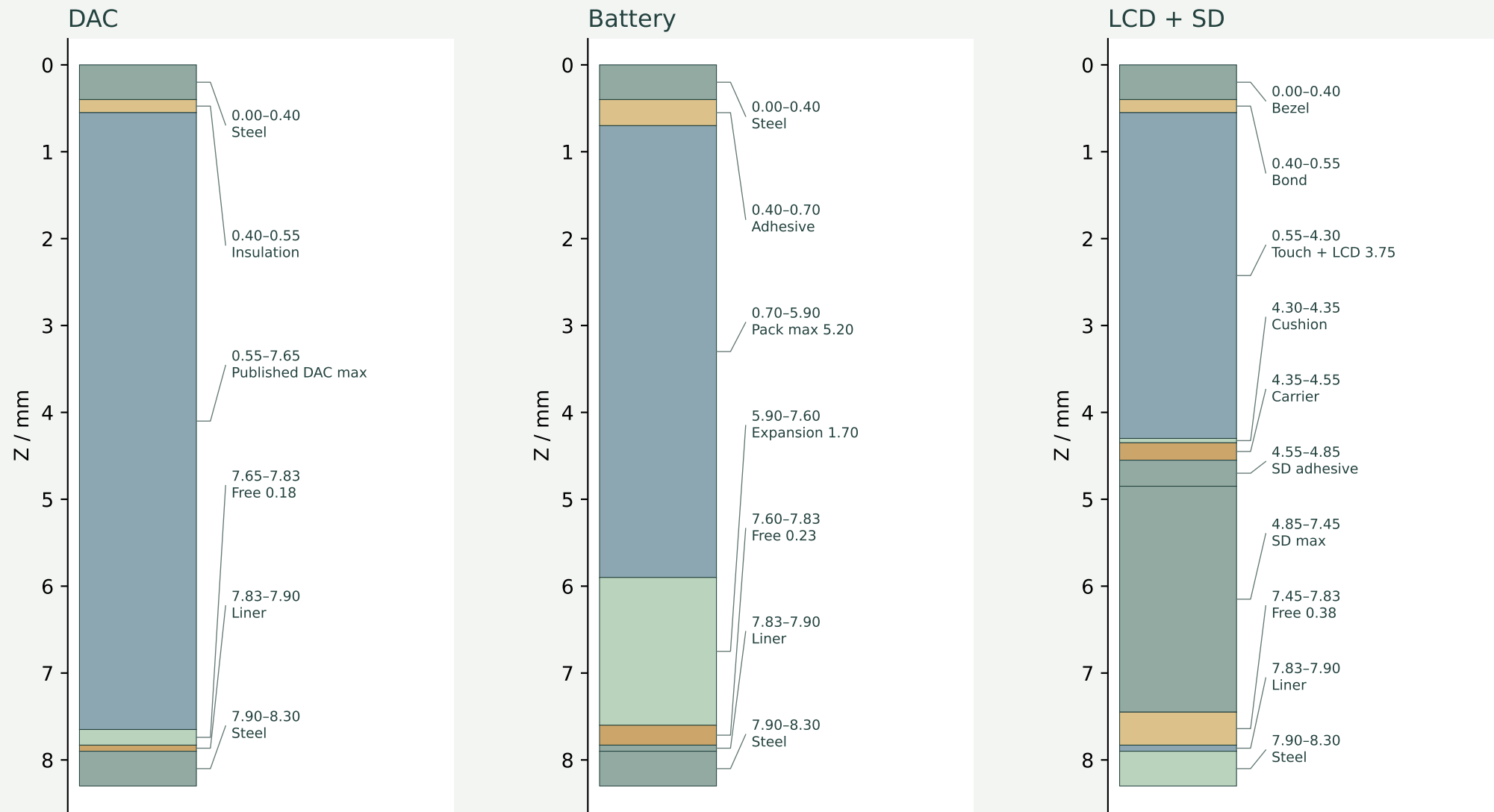


ID	X	Y	Fastener
1	-25.00	61.70	M1.6 × 5 / CSK
2	25.00	61.70	M1.6 × 5 / CSK
3	-29.00	10.00	M1.6 × 5 / CSK
4	29.00	4.00	M1.6 × 5 / CSK
5	29.00	-26.00	M1.6 × 5 / CSK
6	-0.20	-61.00	M1.6 × 5 / CSK
7	-24.00	-61.20	M1.6 × 5 / CSK
C1	-25.00	11.30	M1.4 × 3 / carrier
C2	25.00	11.30	M1.4 × 3 / carrier
C3	-23.00	58.70	M1.4 × 3 / carrier
C4	23.00	58.70	M1.4 × 3 / carrier
X1	22.00	-21.70	M1.4 × 3 / saddle
X2	22.00	1.70	M1.4 × 3 / saddle
D1	24.96	-32.60	M1.4 × 3 / rear column
D2	4.64	-59.27	M1.4 × 3 / rear column
D3	24.96	-59.27	M1.4 × 3 / rear column

Mouth centre	X	Y	Z
usb	31.6000	-9.9927	3.6050
jack	18.8592	-63.6000	4.4225
microsd	-31.6000	29.0000	6.4000

Depth sections and reserves

03 — Layer values are nominal allocations. Full published DAC height is retained.



Manufacturing and assembly checkpoints

04 — STEP defines 3D geometry. DXF profiles are for the named sheet layers only.

Part / check	Specification	Process note
Frame	PA12 / 13 metal threaded inserts	Use BEFORE_HEAT_INSERTS geometry
Front / rear	0.40 mm satin polished stainless	0.03 mm original vector engravings
Carrier	0.20 web / four 0.30 mm welded blades	Weld in flat fixture; inspect distortion
Case threads	7 × PEM MSIB-M1.6-300	Pilot 2.15 / depth 3.77 minimum
Internal threads	6 × PEM MSIB-M1.4-150	Pilot 2.15 / depth 2.27 minimum
XIAO saddle	0.40 steel / 0.30 shield cushion	Two M1.4 × 3 screws; qualify shield load
DAC retention	3 steel columns / insulated seats	Three rear M1.4 × 3 CSK screws
Ports	Actual USB rim + 0.15 radial gap	Positive clearance; optional covers
Release gate	0.18 mm DAC clearance is tight	Tolerance stack and physical cycle tests

Part measurement schedule

05 — Bounding dimensions and assembly position; holes and contours are defined by STEP.

Part	W × L × H	Minimum X, Y, Z
case_insert_1	2.50 × 2.50 × 3.00	-26.25, 60.45, 4.50
case_insert_2	2.50 × 2.50 × 3.00	23.75, 60.45, 4.50
case_insert_3	2.50 × 2.50 × 3.00	-30.25, 8.75, 4.50
case_insert_4	2.50 × 2.50 × 3.00	27.75, 2.75, 4.50
case_insert_5	2.50 × 2.50 × 3.00	27.75, -27.25, 4.50
case_insert_6	2.50 × 2.50 × 3.00	-1.45, -62.25, 4.50
case_insert_7	2.50 × 2.50 × 3.00	-25.25, -62.45, 4.50
carrier_insert_1	2.50 × 2.50 × 1.50	-26.25, 10.05, 2.85
carrier_insert_2	2.50 × 2.50 × 1.50	23.75, 10.05, 2.85
carrier_insert_3	2.50 × 2.50 × 1.50	-24.25, 57.45, 2.85
carrier_insert_4	2.50 × 2.50 × 1.50	21.75, 57.45, 2.85
xiao_clamp_insert_1	2.50 × 2.50 × 1.50	20.75, -22.95, 2.80
xiao_clamp_insert_2	2.50 × 2.50 × 1.50	20.75, 0.45, 2.80
midframe	64.01 × 128.01 × 7.35	-32.00, -64.00, 0.55
front_bezel	64.01 × 128.01 × 0.41	-32.00, -64.00, -0.00
rear_shell	64.01 × 128.01 × 0.41	-32.00, -64.00, 7.90
front_bond	64.01 × 128.01 × 0.16	-32.00, -64.00, 0.40
rear_gasket	64.01 × 128.01 × 0.16	-32.00, -64.00, 7.75
touch_glass	59.96 × 42.62 × 1.25	-29.98, 13.69, 0.55
lcd	60.26 × 42.92 × 2.50	-30.13, 13.54, 1.80

Part measurement schedule

06 — Bounding dimensions and assembly position; holes and contours are defined by STEP.

Part	W × L × H	Minimum X, Y, Z
display_adhesive	59.66 × 42.32 × 0.15	-29.83, 13.84, 0.40
display_carrier	59.00 × 50.00 × 1.45	-29.50, 10.00, 4.35
display_cushion	58.00 × 41.00 × 0.05	-29.00, 14.50, 4.30
audio	25.40 × 33.54 × 6.37	2.10, -63.60, 0.55
xiao	22.48 × 17.78 × 4.46	9.12, -18.89, 0.75
audio_insulator	25.40 × 31.75 × 0.15	2.10, -61.81, 0.40
xiao_insulator	20.90 × 17.70 × 0.35	9.15, -18.85, 0.40
xiao_saddle	3.60 × 27.00 × 0.40	20.20, -23.50, 4.30
xiao_saddle_pad	2.00 × 8.00 × 0.30	21.00, -14.00, 4.00
xiao_clamp_screw_1	2.20 × 2.20 × 3.60	20.90, -22.80, 1.70
xiao_clamp_screw_2	2.20 × 2.20 × 3.60	20.90, 0.60, 1.70
xiao_insertion_stop	0.50 × 15.40 × 1.70	8.40, -16.45, 0.55
xiao_stop_adhesive	0.50 × 15.40 × 0.15	8.40, -16.45, 0.40
dac_seat_1	3.70 × 3.70 × 1.72	23.11, -34.45, 0.55
dac_retainer_1	3.40 × 3.40 × 5.63	23.26, -34.30, 2.27
dac_rear_screw_1	2.81 × 2.81 × 3.01	23.56, -34.01, 5.30
dac_seat_2	3.70 × 3.70 × 1.72	2.79, -61.12, 0.55
dac_retainer_2	3.40 × 3.40 × 5.63	2.94, -60.97, 2.27
dac_rear_screw_2	2.81 × 2.81 × 3.01	3.24, -60.68, 5.30
dac_seat_3	3.70 × 3.70 × 1.72	23.11, -61.12, 0.55

Part measurement schedule

07 — Bounding dimensions and assembly position; holes and contours are defined by STEP.

Part	W × L × H	Minimum X, Y, Z
dac_retainer_3	3.40 × 3.40 × 5.63	23.26, -60.97, 2.27
dac_rear_screw_3	2.81 × 2.81 × 3.01	23.56, -60.68, 5.30
rear_liner	60.40 × 124.40 × 0.07	-30.20, -62.20, 7.83
cell_A_envelope	37.00 × 25.50 × 5.20	-29.00, -18.75, 0.70
cell_A_envelope_adhesive	35.00 × 23.50 × 0.30	-28.00, -17.75, 0.40
cell_B_envelope	25.50 × 37.00 × 5.20	-28.50, -59.00, 0.70
cell_B_envelope_adhesive	23.50 × 35.00 × 0.30	-27.50, -58.00, 0.40
microsd_pcb	22.00 × 18.00 × 0.80	-29.90, 20.00, 4.85
microsd_socket	15.70 × 14.20 × 1.80	-30.85, 21.90, 5.65
microsd_card	15.00 × 11.00 × 1.00	-31.60, 23.50, 5.90
sd_mount_pad	20.00 × 16.00 × 0.30	-28.90, 21.00, 4.55
sd_stop_19_5	20.00 × 0.50 × 0.95	-28.00, 19.25, 4.60
sd_stop_19_5_bond	20.00 × 0.50 × 0.05	-28.00, 19.25, 4.55
sd_stop_38_5	20.00 × 0.50 × 0.95	-28.00, 38.25, 4.60
sd_stop_38_5_bond	20.00 × 0.50 × 0.05	-28.00, 38.25, 4.55
sd_insertion_stop	0.60 × 17.00 × 0.95	-7.60, 20.50, 4.60
sd_insertion_stop_bond	0.60 × 17.00 × 0.05	-7.60, 20.50, 4.55
interface_pcb	15.00 × 32.00 × 0.80	13.50, 23.00, 4.85
interface_pad_24	7.00 × 1.40 × 0.30	14.50, 23.30, 4.55
interface_pad_54	7.00 × 1.40 × 0.30	14.50, 53.30, 4.55

Part measurement schedule

08 — Bounding dimensions and assembly position; holes and contours are defined by STEP.

Part	W × L × H	Minimum X, Y, Z
display_zif	5.00 × 26.50 × 1.10	22.30, 25.75, 5.65
touch_zif	3.50 × 5.00 × 1.10	15.25, 24.50, 5.65
interface_control	6.00 × 12.00 × 1.10	14.00, 35.00, 5.65
pack_protection	24.00 × 3.00 × 1.40	-22.00, 8.50, 0.70
pack_protection_pad	23.00 × 2.50 × 0.30	-21.50, 8.75, 0.40
case_screw_1	3.21 × 3.21 × 5.01	-26.61, 60.09, 3.30
case_screw_2	3.21 × 3.21 × 5.01	23.39, 60.09, 3.30
case_screw_3	3.21 × 3.21 × 5.01	-30.61, 8.39, 3.30
case_screw_4	3.21 × 3.21 × 5.01	27.39, 2.39, 3.30
case_screw_5	3.21 × 3.21 × 5.01	27.39, -27.61, 3.30
case_screw_6	3.21 × 3.21 × 5.01	-1.81, -62.61, 3.30
case_screw_7	3.21 × 3.21 × 5.01	-25.61, -62.81, 3.30
carrier_screw_1	2.20 × 2.20 × 3.60	-26.10, 10.20, 1.55
carrier_screw_2	2.20 × 2.20 × 3.60	23.90, 10.20, 1.55
carrier_screw_3	2.20 × 2.20 × 3.60	-24.10, 57.60, 1.55
carrier_screw_4	2.20 × 2.20 × 3.60	21.90, 57.60, 1.55
usb_cover	0.45 × 12.60 × 6.00	31.55, -16.29, 0.60
sd_cover	0.45 × 13.20 × 2.60	-32.00, 22.40, 5.10
jack_cover	6.50 × 0.45 × 6.50	15.61, -64.00, 1.17